6. Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables (Relevance: Extremely High)

Analysis: This is the most critical first step of this problem. Students must identify the key mathematical variables that affect the "payback time" from a description of shipping costs: such as the fuel unit price (0.42 zeds), annual consumption ($3,500,000$ liters), savings percentage (20%), and equipment cost ($2,500,000$ zeds).

4. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation (Relevance: Extremely High)

Analysis: The problem provides a series of unstructured information (ship specs, fuel prices, estimated savings percentage). Students need to decode this textual information, understand the logical relationships between them, and build an internal model (i.e., Total Cost / Annual Savings Amount = Payback Years) to translate the real-world situation into a mathematical task.

7. Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations (Relevance: High)

Analysis: This aspect is highly relevant as the core task involves handling and processing the numerical data extracted from the text (prices, volumes, percentages, costs) to perform multi-step arithmetic calculations.

7. Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis (Relevance: High)

Correction: The original text provided by the user did not include a number 7 description.
Analysis: Solving this problem requires decomposing the large problem into smaller parts. Students need to first calculate the total annual fuel cost, then the amount saved per year, and finally the number of years required for payback. This ability to break down a complex situation into analyzable steps is central to the solution.

8. Recognise aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures (Relevance: High)

Analysis: Students need to recognize that this is essentially a "percentage application" and a "division/rate" problem. By identifying the problem as a known mathematical concept (e.g., $SavingsAmount = Original\,Expense \times Percentage$), students can recall the appropriate procedures to solve it.

12. Identify the constraints, assumptions simplifications in a mathematical model (Relevance: Medium)

Analysis: While not explicitly requested by the question, as this is an "estimation" problem ("about how many years"), students implicitly make a simplifying assumption when building the model: that annual fuel prices and consumption are constant. Recognizing the need for such simplification is crucial for arriving at a reasonable mathematical result.

9. Translate a problem into a standard mathematical representation or algorithm (Relevance: Medium)

Analysis: Ultimately, students need to translate the logical text into a standard mathematical expression, such as: $Years = 2,500,000/(3,500,000 \times 0.42 \times 0.20)$. This process completes the translation from real-world language to a mathematical algorithm.

1. Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables (相关性：极高)

分析： 这是该题最关键的第一步。学生必须从一段关于航运成本的描述中，识别出哪些是影响“回本时间”的关键数学变量：如燃料单价（0.42 zeds）、年消耗量（3,500,000升）、节省比例（20%）以及设备成本（2,500,000 zeds）。(p. 1)

4. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation (相关性：极高)

分析： 题目提供了一系列非结构化的信息（船舶规格、燃料价格、预估节省比例）。学生需要解码这些文字信息，理解它们之间的逻辑关系，并构建一个内部模型（即：总成本 ÷ 每年节省的金额 = 回本年数），从而将现实情境转化为数学任务。(p. 1)

7. Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis (相关性：高)

分析： 解决该题需要将大问题分解。学生需要先计算每年的总燃料费用，再计算每年能节省多少钱，最后才能计算出年数。这种将复杂情况分解为可分析步骤的能力是解题的核心。(p. 1)

8. Recognise aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures (相关性：高)

分析： 学生需要识别出这本质上是一个“百分比应用”和“除法/比率”问题。通过将问题识别为已知的数学概念（如 $\text{节省金额} = \text{原支出} \times \text{百分比}$），学生可以调用相应的程序来解决它。(p. 1)

12. Identify the constraints, assumptions simplifications in a mathematical model (相关性：中)

分析： 虽然题目没要求，但由于这是一个“估算”问题（"about how many years"），学生在建立模型时实际上隐含了一个简化假设：即每年的燃料价格和消耗量是恒定的。识别出这种简化的需求对于得出合理的数学结果至关重要。(p. 1)

9. Translate a problem into a standard mathematical representation or algorithm (相关性：中)

分析： 最终，学生需要将文本逻辑翻译成标准的数学表达式，例如：$Years = 2,500,000/(3,500,000 \times 0.42 \times 0.20)$。这一过程完成了从现实语言到数学算法的转换。(p. 1)